

David Nguyen

604.307.3521 | dtu7@sfu.ca | [linkedin.com/in/dave-nguyen-53377a268](https://www.linkedin.com/in/dave-nguyen-53377a268) | Burnaby, BC | International Student | Available: 8 Months

Portfolio available: thentd2003.github.io

Experience

Engineering Co-op

Sept. 2023 – Dec. 2024

Analytic Systems

Delta, BC

- Independently designed and simulated (LTspice/Altium) a **high-voltage flyback converter** (24VDC input), my first end-to-end power converter design, sizing the transformer, switching, and snubber components; validated output across an **800–1200V range** against a 1000V target, successfully generating the intended high-voltage surge.
- Designed and built custom **3D-printed test jigs** with integrated PCB slot placement and logically arranged wiring and controls, replacing outdated fixtures; cut per-unit testing time by **20%** and have since been adopted lab-wide, used to test **200+ boards** to date.
- Performed **EMC/EMI compliance testing** (both conducted and radiated) on a batch of **50 power inverters** using a shielded RF enclosure, LISN, and biconical antenna; diagnosed and resolved the sole non-conforming unit, documenting per-unit conformance results for customer-facing QA.
- Enforced strict test-setup discipline across all 50 runs, covering grounding, cable routing, and chamber sealing, holding **100% run accuracy** while cutting per-run setup time by **40%** through optimized fixture layout.
- Calculated individual and parallel-aggregated power ratings for four resistive load units (777.5W each, **3010W combined** at 270V) and **thermally profiled** all six resistor groups under full-load with forced-air cooling, recording peak temperatures up to **210°C** and confirming they stayed within safe operating limits.

Engineering Consultant, Power Electronics

Dec. 2025 – Jul. 2026

Tech Maple Works

Burnaby, BC

- Architected a **half-bridge LCLR series-resonant induction heater**, defining the full power signal chain from DC bus through half-bridge switching, matching inductor, tank capacitor bank, and series-resonant work coil; diagnosed a gate-driver bandwidth limitation causing destructive hard-switching and resolved it via a retargeted 57–70kHz operating band.
- Bench-validated the heater against multiple physical workpieces to confirm magnetic coupling behavior against theory. Ferromagnetic steel coupled and heated efficiently at the design frequency; non-magnetic copper showed minimal heating, consistent with reduced eddy-current coupling.
- Diagnosed Curie-point-driven resonant drift as the open-loop design's core fragility and architected a **four-stage closed-loop PLL** (tank-voltage feedback, phase detector/VCO, gate-driver stage) to track resonance in real time.
- Currently designing custom **PCBs** for the full device, covering the power stage, PLL feedback, and gate driver circuitry, replacing the breadboard prototype with a production-ready layout.

Projects

Embedded Systems & Mechatronics: ESP32 Pan-Tilt Soccer/Ball Tracking Camera

[view full build →](#)

- Own the embedded/firmware side of a pan-tilt ball-tracking camera: implemented a **BLE peripheral (GATT server)** on an **ESP32**, architecting a non-blocking firmware state machine for tracking, homing, and limit enforcement.
- Diagnosed asymmetric CW/CCW rotation speeds in the continuous-rotation servo and built a **dead-reckoned software position encoder** in C++ (no physical encoder hardware), with separate per-direction calibration models tuning reliable pulse timing down to **130ms** for precise positional tracking.
- Produced a **BLE protocol reference document** to hand off the peripheral-side implementation, enabling a teammate to build the iOS **BLE central** app and on-device AI ball-detection logic in parallel.
- Designed a bearing-mounted rotating pan base using a **1.5:1 bevel-gear** reduction (module 1.5, 45T fixed / 30T pinion) reverse-engineered around a non-co-located motor layout; diagnosed gear-mesh flex risk and resolved it by supporting both gear faces near their bearings, calculating bore tolerances (nominal OD + 0.53–0.6× nozzle diameter) for a reliable **3D-printed** fit, modeled in **SolidWorks**.

Education

Simon Fraser University

2021 – Present

Bachelor of Applied Science in Engineering Science (Electronics)

Burnaby, BC

Simon Fraser University

May 2023 – Sep 2023

Siemens Mechatronic Systems Certification Program (Level 1)

Surrey, BC

Skills & Interests

Hardware & Lab: Soldering, Oscilloscope & DMM, PCB Design & Fabrication, **3D Printing**, Circuit Troubleshooting

Design & Simulation: Altium Designer, LTspice, KiCad, OrCAD, Proteus, MATLAB/Simulink, SolidWorks

Programming & HDL: C++, Python, VHDL, Verilog, SystemVerilog, Xilinx Vivado, Intel Quartus, ModelSim

Languages: English (Proficient), Vietnamese (Native)